

UID EX 10

Name : Madhumitha P

Reg.no : 240701297

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File

```
<> INDEX.HTML > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Student Performance Visualization</title>
7
8  <style>
9  body {
10     font-family: Arial;
11     text-align: center;
12     margin: 50px;
13  }
14  canvas {
15     margin: 20px auto;
16  }
17 </style>
18
19 </head>
20 <body>
21
22 <h1>Student Performance System</h1>
23
24 <canvas id="pieChart" width="400" height="400"></canvas>
25 <canvas id="barChart" width="400" height="400"></canvas>
26
27 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
28 <script src="script.js"></script>
29
30 </body>
31 </html>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

JS script.js > ...
1  // Data for student performance
2  const studentData = {
3    labels: ['Math', 'Science', 'English', 'Computer', 'History'],
4    datasets: [{
5      label: 'Marks Obtained',
6      data: [85, 90, 75, 95, 80],
7      backgroundColor: [
8        '#FF6384',
9        '#36A2EB',
10       '#FFCE56',
11       '#4BC0C0',
12       '#9966FF'
13     ]
14   }]
15 };
16
17 // PIE CHART (Subject-wise distribution)
18 const ctxPie = document.getElementById('pieChart').getContext('2d');
19
20 new Chart(ctxPie, {
21   type: 'pie',
22   data: studentData,
23   options: {
24     responsive: true,
25     plugins: {
26       title: {
27         display: true,
28         text: 'Marks Distribution by Subject'
29       }
30     }
31   }
32 });
33
34 // BAR CHART (Marks comparison)
35 const ctxBar = document.getElementById('barChart').getContext('2d');
36
37 new Chart(ctxBar, {
38   type: 'bar',
39   data: studentData,
40   options: {
41     responsive: true,
42     plugins: {
43       title: {
44         display: true,
45         text: 'Marks Comparison'
46       }
47     },
48     scales: {
49       y: {
50         beginAtZero: true
51       }
52     }
53   }
54 });

```

OUTPUT:

Student Performance System

